

HYDRATE LESS, RENDER MORE WITH NUXT

enterJS

Denis Lugowski

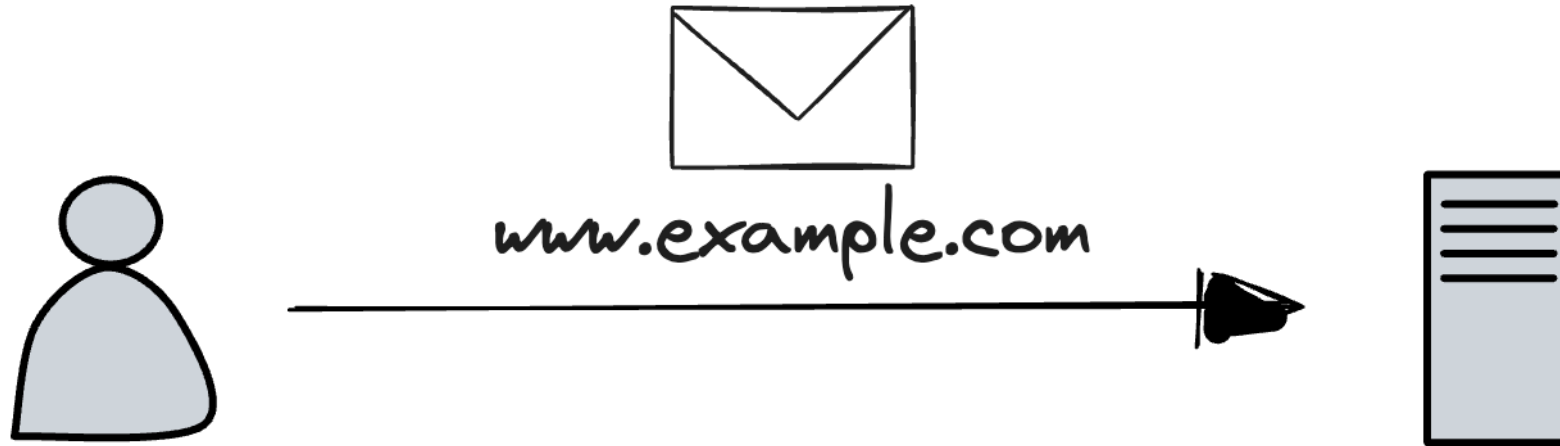
Senior Full Stack Developer 🦄

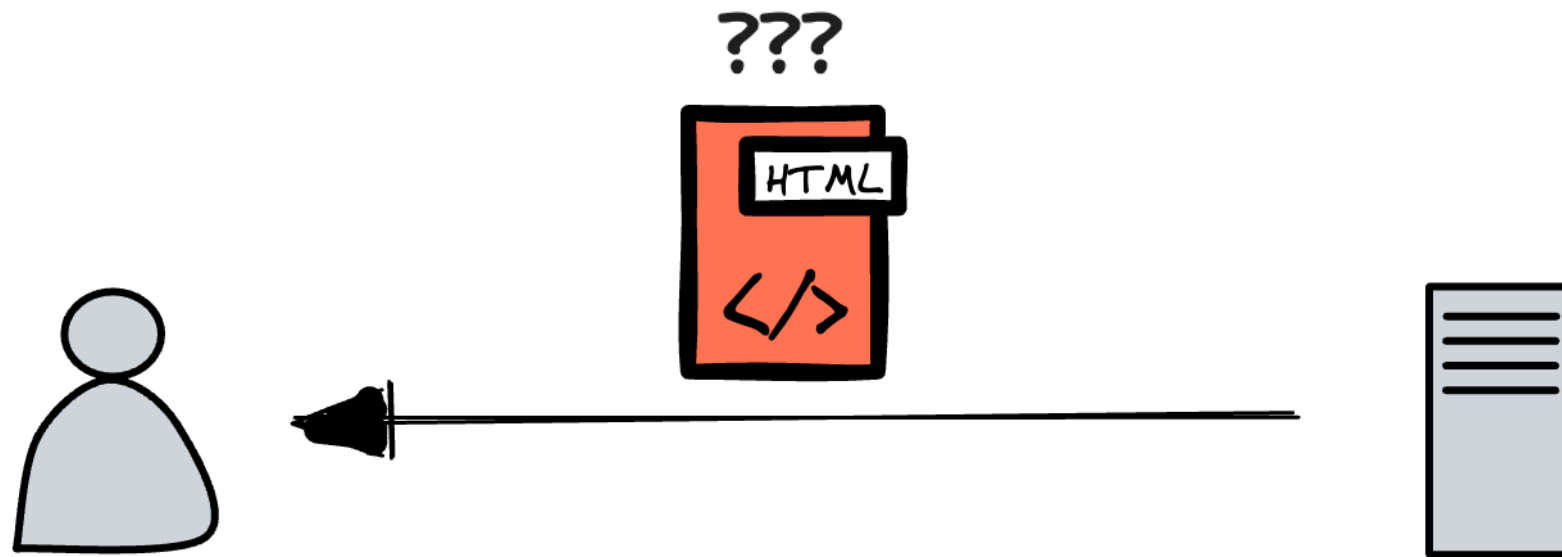
- Started career in low-level programming
- 5 years of Vue/Nuxt experience
- Since Jan 2026 organizer of the Vue.js Hamburg Meetup 



ISLANDS







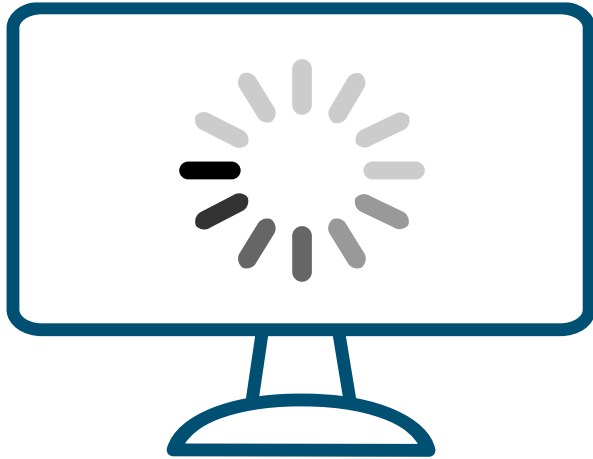
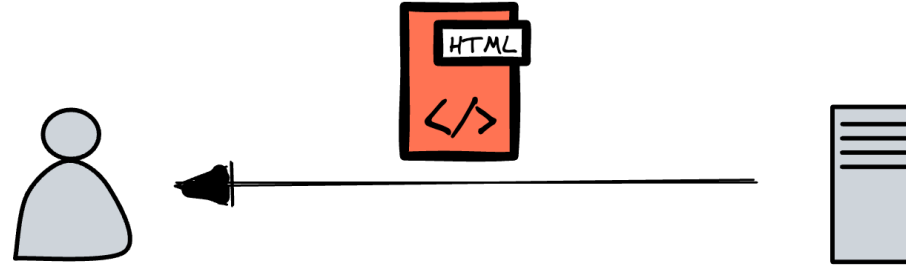


```
CSR example

<!DOCTYPE html>
<html>
  <head>
    <title>Nuxt Rendering Modes Demo</title>
    <script type="module" src="/_nuxt/B_5vHM_Q.js" crossorigin></script>
  </head>
  <body>
    <div id="__nuxt"></div>
  </body>
</html>
```

```
SSR example

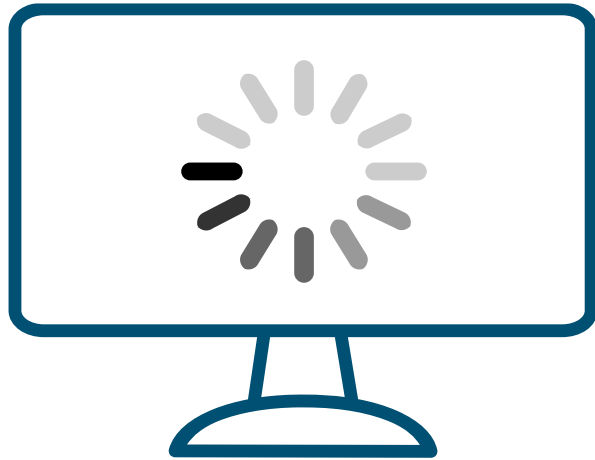
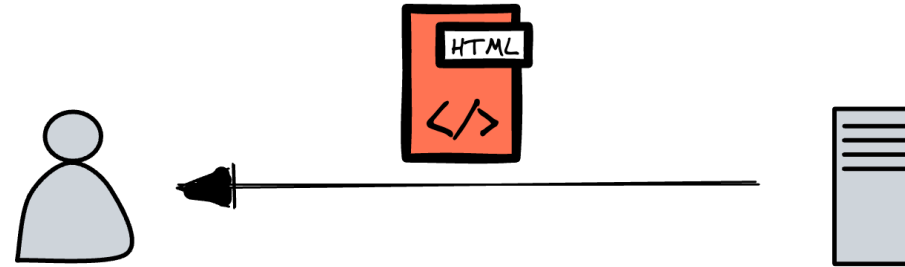
<!DOCTYPE html>
<html>
  <head>
    <title>Nuxt Rendering Modes Demo</title>
    <script type="module" src="/_nuxt/B_5vHM_Q.js" crossorigin></script>
  </head>
  <body>
    <div id="__nuxt">
      <div>
        <header class="header">
          <div class="header-content">
            <h1 class="logo">
              <a href="/"> Nuxt Render Modes Demo</a>
            </h1>
            <nav class="nav">
              <a href="/" class="nav-link">Home</a>
              <a href="/ssr" class="active nav-link">SSR</a>
              ...
            </nav>
          </div>
        </header>
        <main class="container">
          ...
        </main>
      </div>
    </div>
  </body>
</html>
```



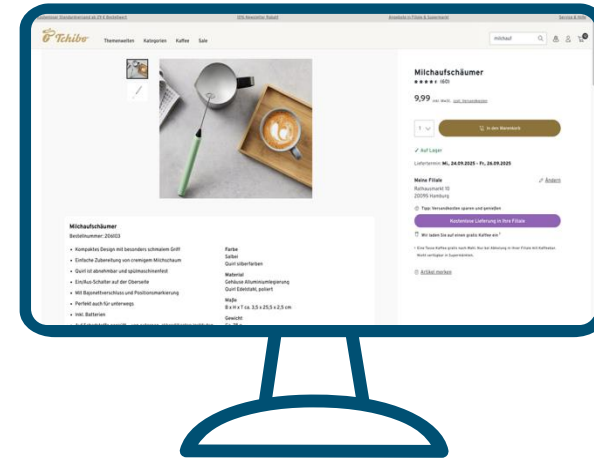
Client-side rendering (CSR)

```
SSR example

<!DOCTYPE html>
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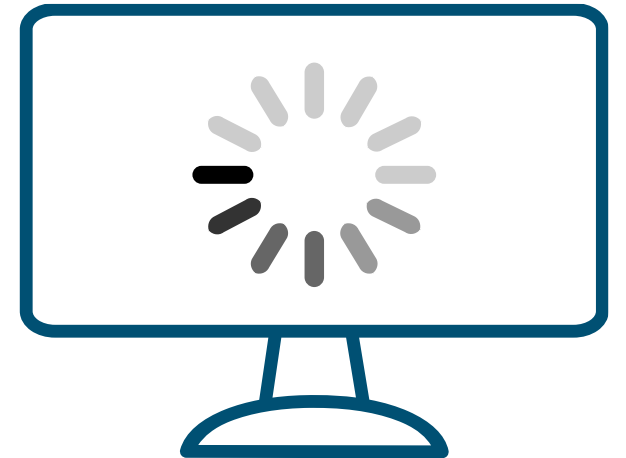
Client-side rendering (CSR)



Server-side rendering (SSR)

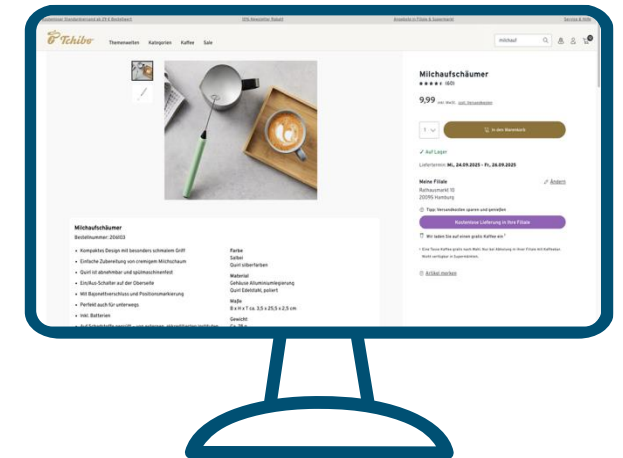
CLIENT-SIDE RENDERING (CSR)

- Renders content only in browser
- Suitable for highly interactive pages
- Initial load takes some time
- HTML content not included in first response



SERVER-SIDE RENDERING (SSR)

- Page is rendered on the server-side
- HTML content delivered in first response
- Fast initial content
- Hydration phase for interactivity



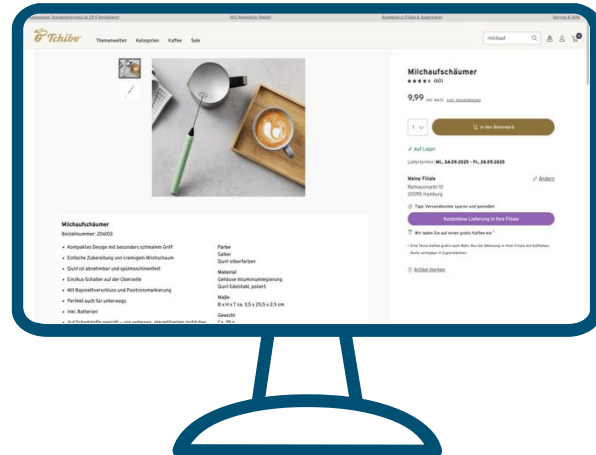
STATIC SITE GENERATION (SSG)

- Rendering at build time instead of per request (as in SSR)
- Suitable for less frequently changed content
- Can contain interactive content

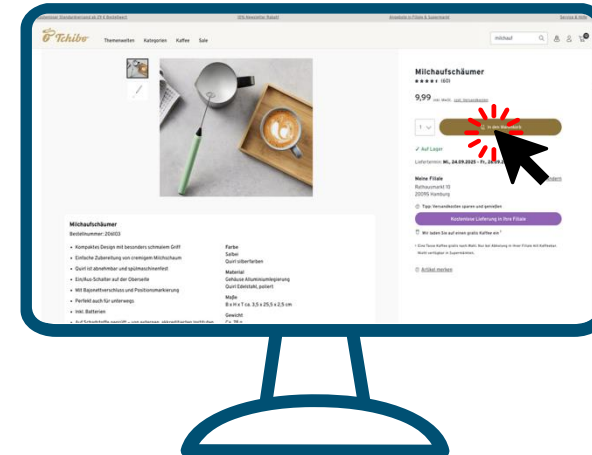


HYDRATION

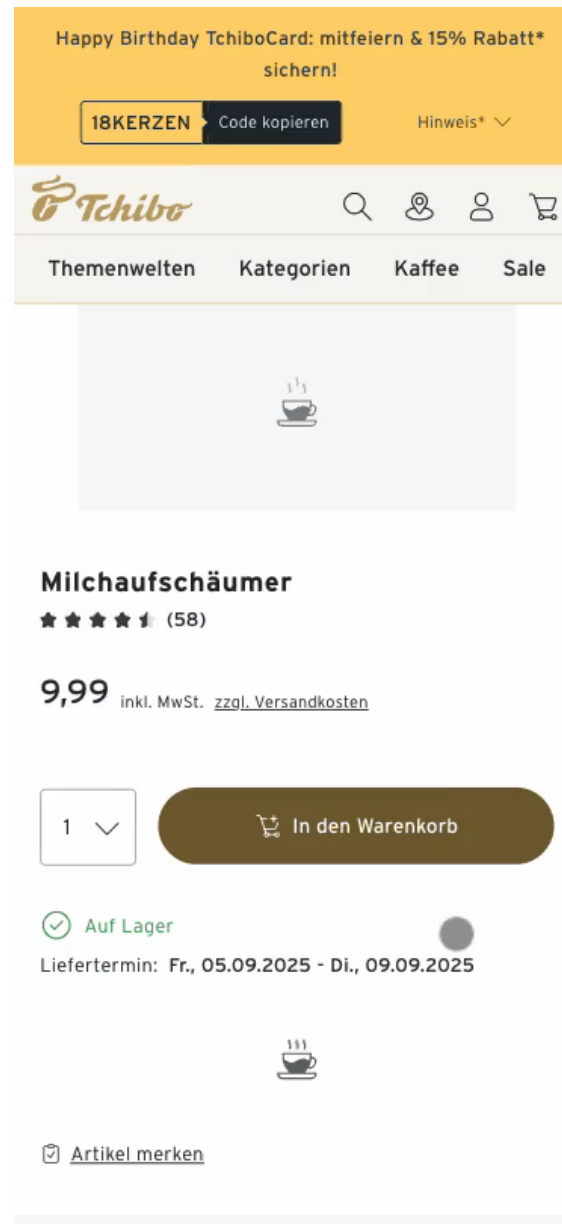
1. Match HTML elements to Vue components
2. Attach event listeners to HTML elements



Static



Interactive



Test settings:

6x CPU slowdown

Slow 4G network

RAGE CLICK DEMO

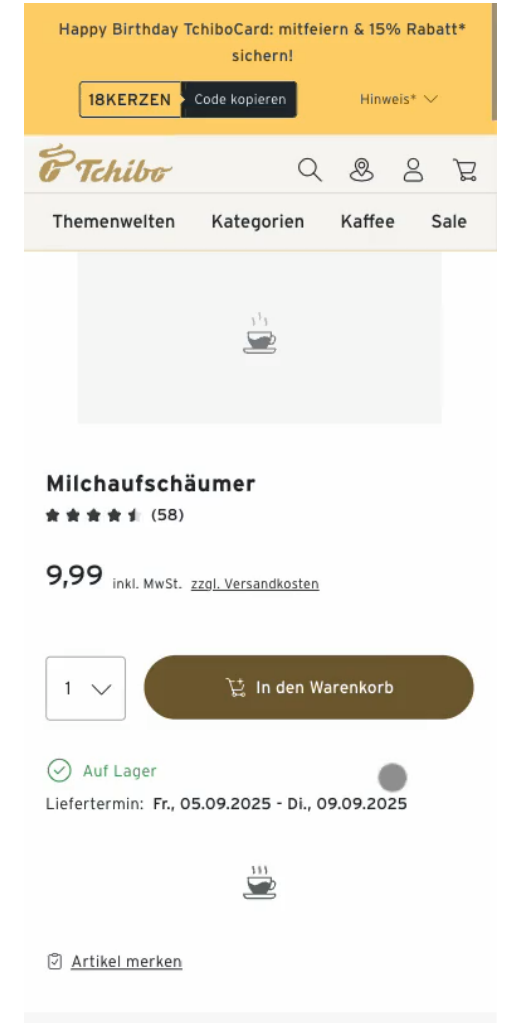
- Problem of every hydrated application
- Hydration time can't be eliminated
- Strategies



No hydration



Delay hydration



ISLANDS

- Static content sprinkled with interactive islands
- Hydration prioritization of islands
- Reduced likelihood for users to experience rage clicks

NUXT ISLANDS

- Interactive content with mostly non-interactive islands
 - Can still be interactive using props with one caveat
- Content rendered fully on the server-side
- Reduced hydration time, smaller bundle size



NUXT LAZY HYDRATION

- Not a rendering mode, but an optimization technique
- Defer hydration of non-critical interactive components
- Can be applied to SSR, SSG



DEMO TIME



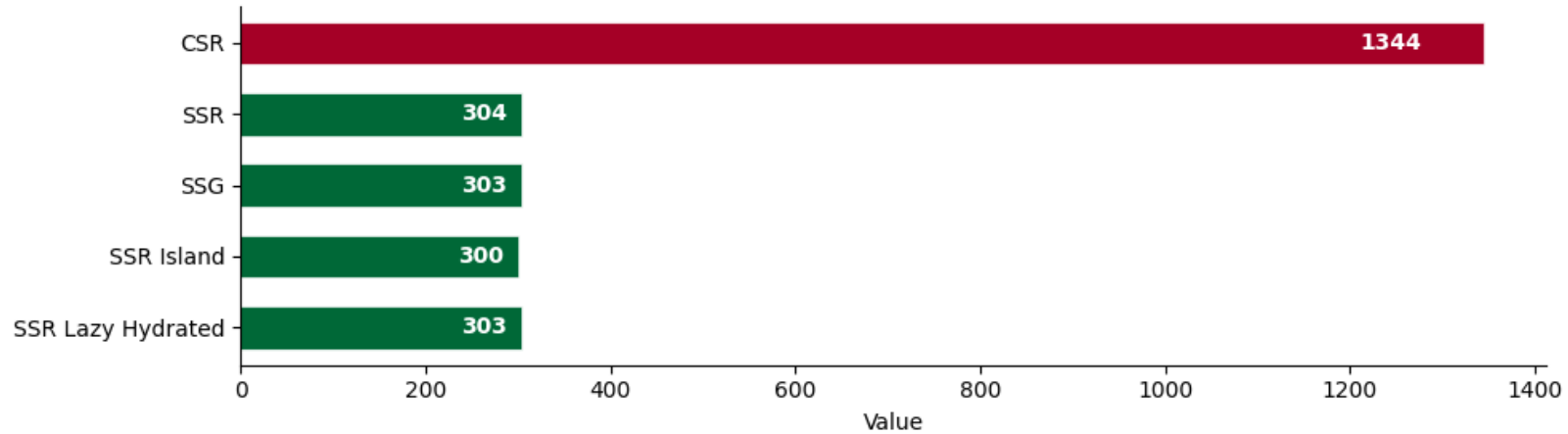
<https://blog.lancedb.com/guide-to-use-contextual-retrieval-and-prompt-caching-with-lancedb/>

COMPARISON OF RENDERING MODES

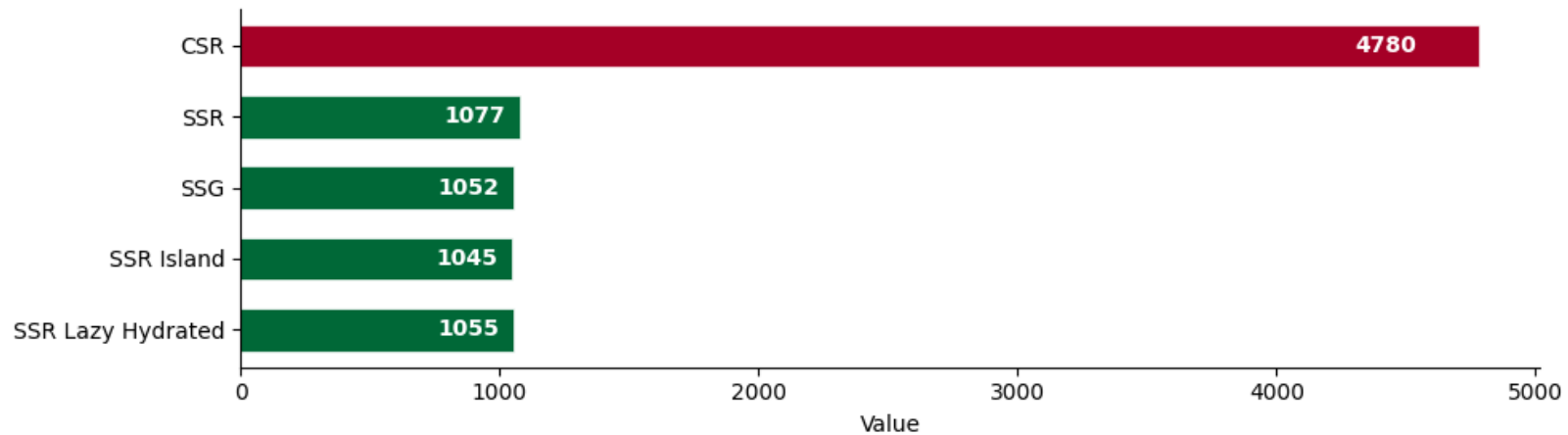
- Demo Nuxt 3 application
- Page load test (10 runs, p75, medium/high throttling)
- Criteria
 - First Contentful Paint (FCP)
 - Hydration Time
 - Client Bundle Size

FCP – p75, 10 runs (ms)

Medium throttling

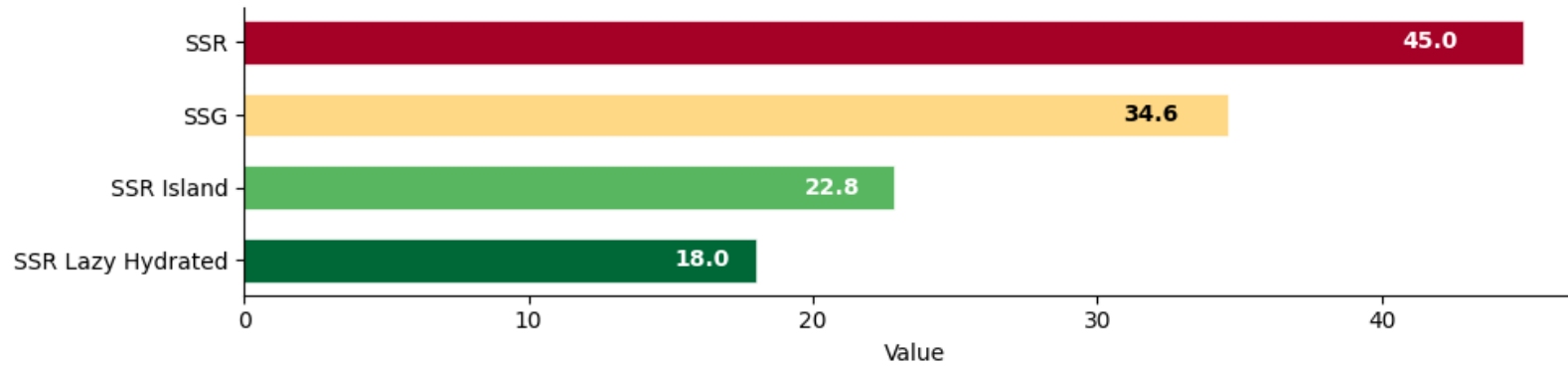


High throttling

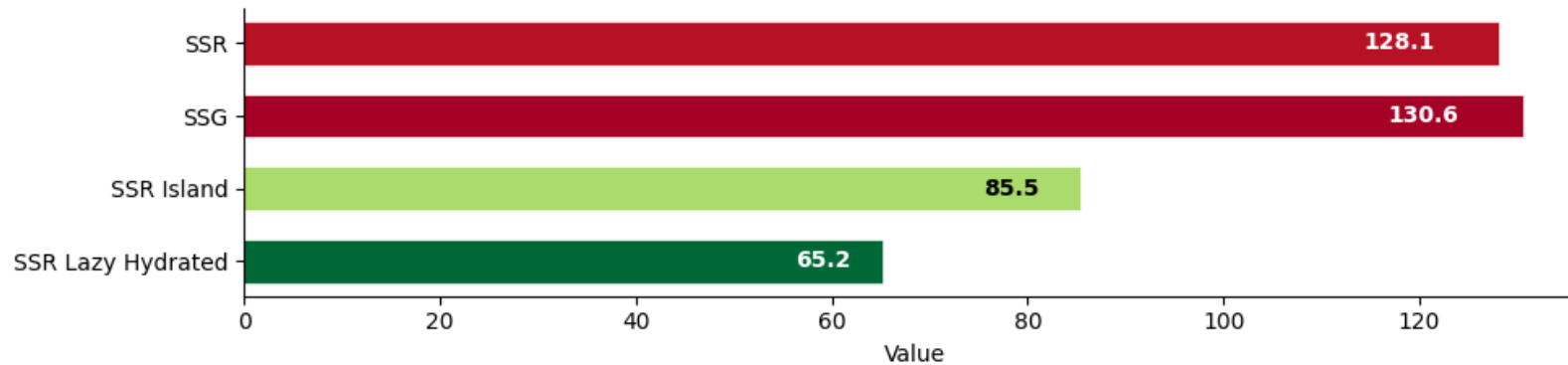


Hydration Time – p75, 10 runs (ms)

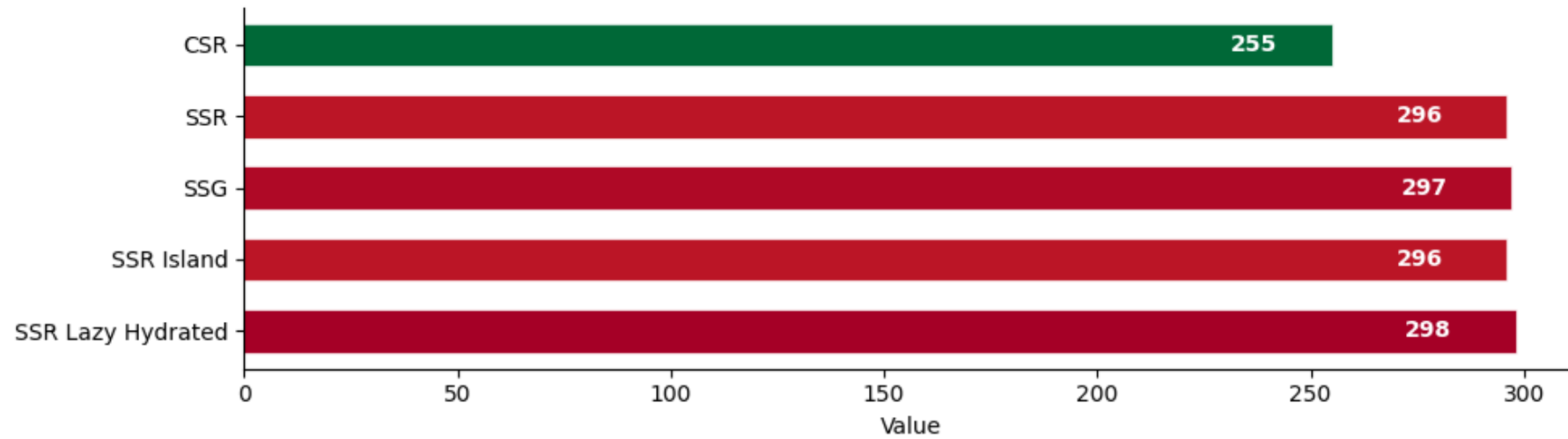
Medium throttling



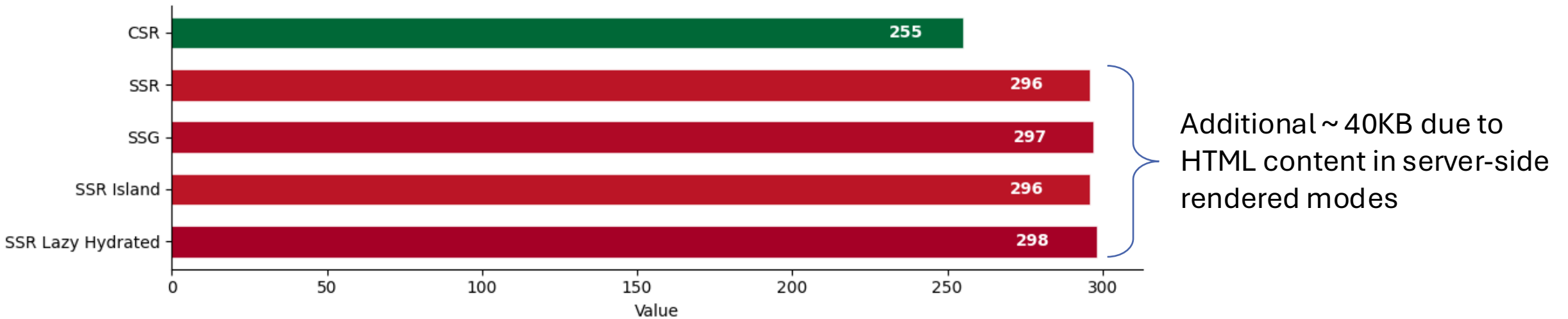
High throttling



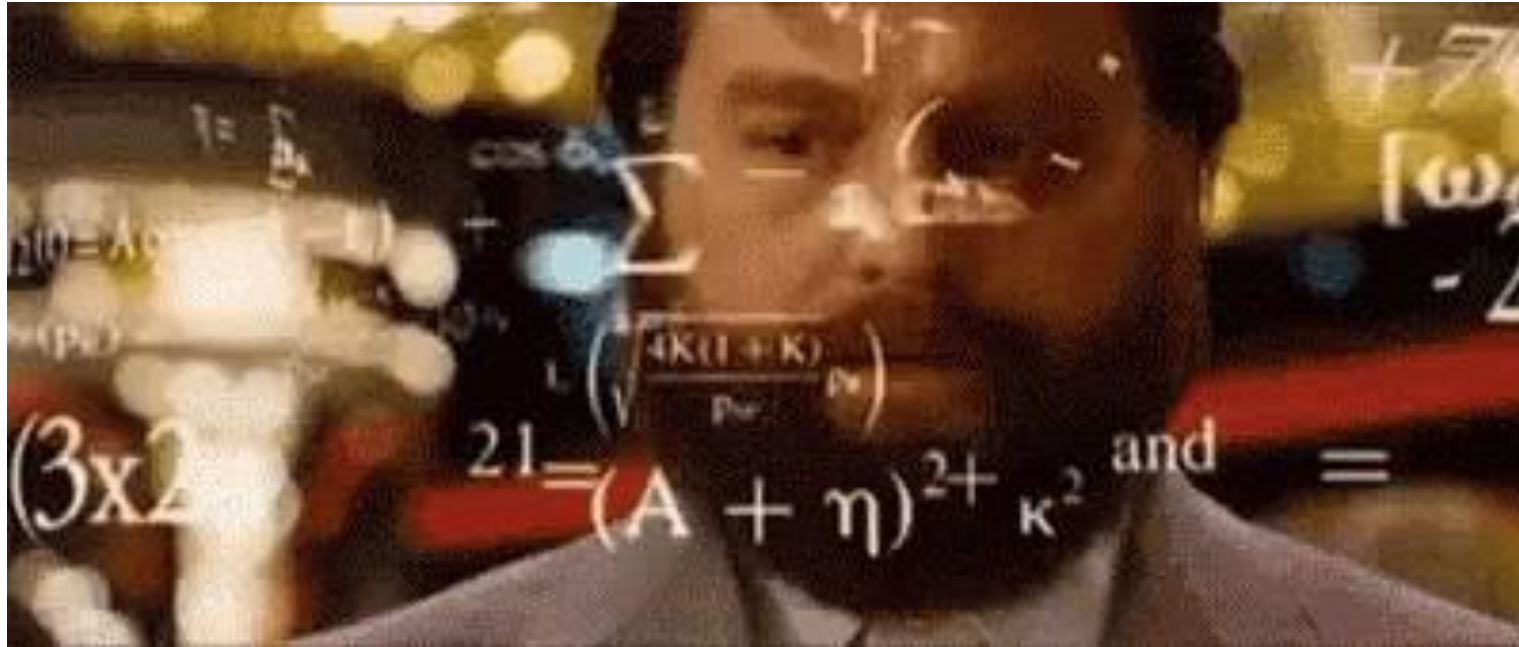
Bundle Size (KB)



Bundle Size (KB)



WHICH IS THE RIGHT RENDERING MODE FOR MY APP?



PICK THE RIGHT RENDERING MODE

Use Case	CSR	SSR	SSG	Nuxt Islands	Lazy Hydration
Documentation, Blog, Landing page					
E-commerce					
Backoffice, Web editor, Game etc.					

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PICK THE RIGHT RENDERING MODE

Use Case	CSR	SSR	SSG	Nuxt Islands	Lazy Hydration
Documentation, Blog, Landing page	✗	✓	🏆	🟡	🟡
E-commerce	✗	🏆	✗	✓	✓
Backoffice, Web editor, Game etc.					

PICK THE RIGHT RENDERING MODE

Use Case	CSR	SSR	SSG	Nuxt Islands	Lazy Hydration
Documentation, Blog, Landing page	✗	✓	🏆	●	●
E-commerce	✗	🏆	✗	✓	✓
Backoffice, Web editor, Game etc.	🏆	✓	✗	✗	✗

Always Benchmark



Consider islands

Identify lazy-loadable parts

What is the purpose of the page?

THANK YOU!

*Check out the
code here:*

